Unit-3
Lecture-3

Position, Momentum and Energy operators

① Position operator

$\hat{x}\psi = x\psi$

$\boxed{\hat{x} = x}$ $\quad \hat{y} = y \quad \hat{z} = z$

② Momentum operator

$\hat{P}_x = -i\hbar \frac{\partial}{\partial x}$

$\hat{P} = -i\hbar \vec{\nabla}$

③ Total energy operator

$\hat{E} = i\hbar \frac{\partial}{\partial t}$, $\quad \hat{H} = \frac{-\hbar^2}{2m}\nabla^2 + V$

②

$\psi = Ae^{i(kx - \omega t)}$

$K = \frac{P_x}{\hbar} \qquad \omega = \frac{E}{\hbar}$

$\psi = Ae^{\frac{i}{\hbar}(P_x x - Et)}$

$\frac{d\psi}{dx} = Ae^{\frac{i}{\hbar}(P_x x - Et)} \cdot \frac{P_x i}{\hbar}$

$\frac{d\psi}{dx} = \frac{P_x}{\hbar} i \psi$

$\frac{i \times \hbar}{i \times i} \frac{\partial \psi}{\partial x} = \psi P_x$

$\frac{i\hbar}{i^2} \frac{\partial \psi}{\partial x} = P_x \psi$

$-i\hbar \frac{\partial}{\partial x} = \hat{P}_x$

$\hat{P}_y = -i\hbar \left(\frac{\partial}{\partial y}\right)$

$\hat{P} = -i\hbar \underbrace{\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)}_{\vec{\nabla}}$

3(a)

$$\psi = Ae^{\frac{i}{\hbar}(p_x x - Et)}$$

$$\frac{\partial \psi}{\partial t} = Ae^{\frac{i}{\hbar}(p_x x - Et)} \cdot -\frac{Ei}{\hbar}$$

$$-\frac{\hbar}{i}\frac{\partial \psi}{\partial t} = \psi E$$

$$\frac{-\hbar \times i}{i \times i}\frac{\partial \psi}{\partial t} = \psi E$$

$$\boxed{i\hbar\frac{\partial}{\partial t} = \hat{G}}$$

3(b)

$$T = E_K + V$$

$$T = \frac{p^2}{2m} + V$$

$$T = \frac{\vec{p} \cdot \vec{p}}{2m} + V$$

[T=H]

$$\hat{H} = \frac{-i\hbar\vec{\nabla} \cdot -i\hbar\vec{\nabla}}{2m} + V$$

$$\hat{H} = \frac{-\hbar^2\vec{\nabla}^2}{2m} + V$$

ψ

Eigen values and Eigen functions

If an operator $\hat{A}$ acts on a well defined wave function ψ and the result is the same wave function $\psi(x)$ multiplied by constant 'a' then 'a' is the eigen value of eigen function $\psi(x)$

$$\underbrace{\hat{A}\,\Psi(x)}_{\text{eigen function}} = \underbrace{a}_{\text{eigen value}}\,\underbrace{\Psi(x)}_{\text{eigen function}}$$

① $\psi(x) = e^{-5x}$, $\hat{A} = \frac{d}{dx}$

② $\psi(x) = \sin 2x$, $\hat{A} = \frac{d^2}{dx^2}$

③ $\psi(x) = e^{-x}$, $\hat{A} = \frac{d}{dx}$

④ $\psi(x) = \cos x$, $\hat{A} = \frac{d}{dx}$

① $\frac{d}{dx} e^{-5x}$

$\hat{A}\psi(x) = \psi(x)\,a$

$\frac{d}{dx} e^{-5x}$

$$= \underbrace{-5}_{\text{eigen value}}\;\underbrace{e^{-5x}}_{\text{eigen function}}$$

(ii) $\frac{d^2}{dx^2} \sin 2x$

$\frac{d}{dx}(2\cos 2x)$

$2 \cdot 2(-\sin 2x)$

$$\underbrace{-4}_{\text{eigen value}}\;\underbrace{\sin 2x}_{\text{eigen function}}$$